Tutorial 4 (Test Review)

Wednesday, February 5, 2025 7:10 PM

PLAN

Practice Test 1
Q2,20,19,17,14,12,10,8

For my 1A43 tutorial I will try practice test 2.
Liam (other TA) is doing:
Sample 1: 4,14,16,17
Sample 2: 4,15,19,20

2. Which of the following series converge?

(i) $\sum_{n=1}^{\infty}\frac{e^{-\sqrt{n}}}{\sqrt{n}}$ (ii) $\sum_{n=2}^{\infty}\frac{1}{n(\ln n)^3}$

(a) (i) only **(b)** (ii) only **(c)** (i) and (ii) **(d)** neither

check (i): try integral test:

$\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}dx$ $u=-x^{1/2}$ $du=\frac{-1}{2}x^{-1/2}dx \Rightarrow -2du=\frac{1}{\sqrt{x}}dx$

as $x\to1$ $u\to-1$
$x\to\infty$ $u\to-\infty$

$=\int_{-1}^{-\infty}(-2)e^u du$

$= -2e^u+c\Big|_{-1}^{-\infty} = \lim_{u\to-\infty}\overset{0}{\cancel{-2e^u}} + 2e^{-1}$ convergent.

Check (ii): try integral test again:

$\int_2^\infty \frac{1}{x(\ln(x))^3}dx$ $u=\ln(x)$ $du=\frac{1}{x}dx$

as $x\to2$ $u\to\ln(2)$
$x\to\infty$ $u\to\infty$

$=\int_{\ln(2)}^{\infty}u^{-3}du$

$=\frac{u^{-2}}{-2}+c\Big|_{\ln(2)}^{\infty} = \left(\lim_{u\to\infty}\overset{0}{\cancel{\frac{-1}{2u^2}}}\right)+\frac{1}{2(\ln(2))^2}$ converges.

ANSWER: C (i) and (ii) converge.

20. Find the radius of convergence of the following power series.

$$\sum_{n=0}^{\infty}\underbrace{(-1)^n}_{\text{no need to worry for abs. value.}}\frac{1\cdot4\cdot7\cdot\dots\cdot(3n-2)}{n!3^n}x^{3n}$$

$3n+3-2$

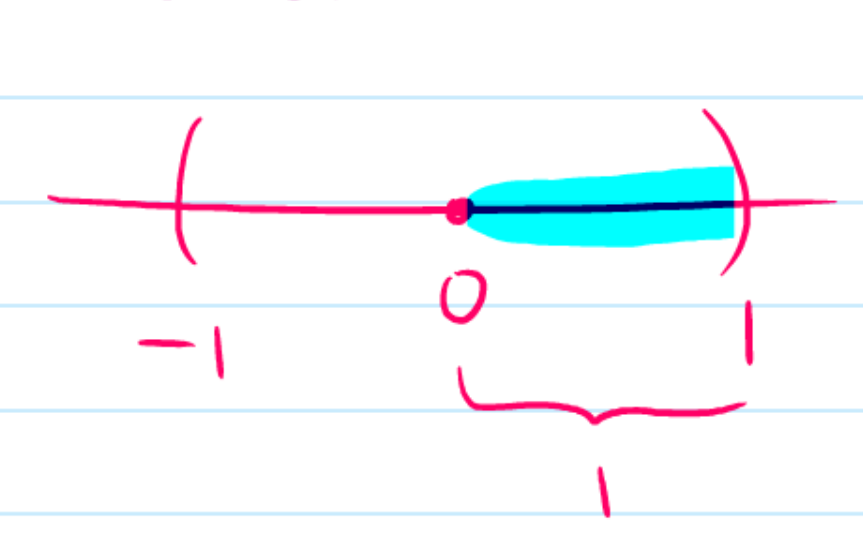
(a) $\sqrt[3]{3}$ **(b)** $\frac{1}{\sqrt[3]{3}}$ **(c)** 1 **(d)** 0 **(e)** ∞

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\cancel{1\cdot4\cdot7}\cdots\cancel{(3n-2)}(3n+1)}{(n+1)!\,3^{\cancel{n}+1}}x^{\cancel{3n}+3}\cdot\frac{n!\,\cancel{3^n}}{\cancel{x^{3n}}\cdot\cancel{1\cdot4\cdot7}\cdots\cancel{(3n-2)}}\right|$$

$$=\left|\frac{(3n+1)}{3(n+1)!}n!\,x^3\right| = \left|\frac{(3n+1)}{3}\frac{\cancel{n!}}{(n+1)\cancel{(n)!}}x^3\right|$$

Now, $\lim_{n\to\infty}\frac{(3n+1)}{3(n+1)}|x^3| = |x^3|<1$

Hence $-1<x^3<1 \Rightarrow (-1)^{1/3}<x<(1)^{1/3} \Rightarrow -1<x<1$



the radius of convergence is 1. (c)

19. Find the radius of convergence of

$$\sum_{n=1}^{\infty}(n+1)!(3x-1)^n$$

(a) ∞ **(b)** 0 **(c)** 1 **(d)** $\frac{1}{3}$ **(e)** $\frac{2}{3}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+2)!(3x-1)^{n+1}}{(n+1)!(3x-1)^n}\right| = \left|\frac{(n+2)\cancel{(n+1)!}}{\cancel{(n+1)!}}(3x-1)\right| = (n+2)|3x-1|$$

$\lim_{n\to\infty}(n+2)|3x-1| = \infty > 1$ this diverges! !!

NO RADIUS! Hence $\underline{0}$ ⇒ (b).

17. Find the interval of convergence of

$$\sum_{n=1}^{\infty}\frac{n(3x+2)^n}{n^2+1}$$

(a) $(-\infty,\infty)$ **(b)** $[-\frac{1}{3},-\frac{2}{3}]$ **(c)** $[-\frac{1}{3},-\frac{2}{3})$ **(d)** $[-1,-\frac{1}{3}]$ **(e)** $[-1,-\frac{1}{3})$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)(3x+2)^{n+1}}{(\underbrace{(n+1)^2}_{n^2+2n+1}+1)}\times\frac{(n^2+1)}{n(3x+2)^n}\right| = \frac{(n+1)}{n}\frac{(n^2+1)}{[n^2+2n+2]}|3x+2|$$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\underbrace{\frac{(n+1)}{n}}_{\approx1}\underbrace{\frac{(n^2+1)}{(n^2+2n+2)}}_{\approx1}|3x+2| = |3x+2|<1$$

$\Rightarrow -1<3x+2<1 \Rightarrow \frac{-3}{3}<\frac{3x}{3}<\frac{-1}{3} \Rightarrow -1<x<\frac{-1}{3}$

Need to check endpoints now.

When $x=-1$: $(3x+2)=(-3+2)=(-1)$

So $\sum_{n=1}^{\infty}\frac{n(-1)^n}{n^2+1}$ check AST: $b_n=\frac{n}{n^2+1}$

$f(x)=\frac{x}{x^2+1}$ $f'(x)=\frac{(x^2+1)-x(2x)}{(x^2+1)^2}=\frac{-x^2+1}{(x^2+1)^2}<0$ for $x>1$

So b_n is decr and $\lim_{n\to\infty}\frac{n}{n^2+1}=0$

By AST: converges at $x=-1$.

When $x=-1/3$: $(3x+2)=(-1+2)=1$

So $\sum_{n=1}^{\infty}\frac{n}{n^2+1}$ note: $\frac{n}{n^2+1}>\frac{n}{n^2+n^2}=\frac{n}{2n^2}=\frac{1}{2n}$ ↗ p-series $p=1$ diverges

by comparison test it diverges at $x=-1/3$.

ANS: $[-1,-1/3)$ (e)

14. What is the minimum number of terms needed in order to estimate the following sum to within .001?

$$\sum_{n=0}^{\infty}\frac{(-1)^n}{(n+1)!}$$

(a) 2 **(b)** 3 **(c)** 4 [✓ **(d)** 5] **(e)** 6

if you have $(-1)^n$ try Alternating Series Estimation Thm:

If $\sum_{n=1}^{\infty}(-1)^n b_n$ converges by AST then $|S-S_n|\le b_{n+1}$

GOAL: test values for b_n. $b_n=\frac{1}{(n+1)!}$ Compare: $0.001=\frac{1}{1000}$

$b_1=\frac{1}{2!}=\frac{1}{2}>\frac{1}{1000}$

$b_2=\frac{1}{3!}=\frac{1}{6}>\frac{1}{1000}$

$b_3=\frac{1}{4!}=\frac{1}{24}>\frac{1}{1000}$

$b_4=\frac{1}{5!}=\frac{1}{120}>\frac{1}{1000}$

$b_5=\frac{1}{6!}=\frac{1}{720}>\frac{1}{1000}$

$b_6=\frac{1}{7!}=\frac{1}{5040}<\frac{1}{1000}$ FINALLY!

$7(700+20)=4900+140$

By thm, $|S-S_5|\le b_6<\frac{1}{1000}$

So: min terms needed: ⑤.

12. If the n^{th} partial sum of the series $\sum_{n=0}^{\infty}a_n$ is $s_n=\frac{2n+1}{4n+3}-\frac{n}{\ln n}$, find $\sum_{n=0}^{\infty}a_n$.

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$ [✓ **(e)** divergent]

Recall: $\sum_{n=0}^{\infty}a_n=\lim_{N\to\infty}\sum_{n=0}^{N}a_n=\lim_{N\to\infty}S_N$

So, $=\lim_{n\to\infty}\frac{2n+1}{4n+3}-\frac{n}{\ln(n)} = \underbrace{\lim_{n\to\infty}\frac{2n+1}{4n+3}}_{1/2}-\underbrace{\lim_{n\to\infty}\frac{n}{\ln(n)}}_{\text{use functions.}}$

$f(x)=\frac{x}{\ln(x)}$ $\lim_{x\to\infty}\frac{x}{\ln(x)}\overset{l'h}{=}\lim_{x\to\infty}\frac{1}{1/x}=\lim_{x\to\infty}x$ diverges.

So, this diverges.

10. If you were to use use Mathematical Induction to show that

$$\sum_{i=1}^{n}i^2=\frac{n(n+1)(2n+1)}{6},$$

which of the following would be the second step?

(a) Assume $\sum_{i=1}^{k}i^2=\frac{k(k+1)(2k+1)}{6}$, and show that $\sum_{i=1}^{k}(i+1)^2=\cancel{\frac{(k+1)(k+2)(2k+3)}{6}}$

(b) Assume $\sum_{i=1}^{k}i^2=\frac{k(k+1)(2k+1)}{6}$, and show that $\sum_{i=1}^{k+1}i^2=\frac{(k-1)k(2k-1)}{6}$

(c) Assume $\sum_{i=1}^{k}i^2=\frac{k(k+1)(2k+1)}{6}$, and show that $\sum_{i=1}^{k-1}i^2=\frac{(k-1)k(2k-1)}{6}$

[✓ **(d)**] Assume $\sum_{i=1}^{k}i^2=\frac{k(k+1)(2k+1)}{6}$, and show that $\boxed{\sum_{i=1}^{k+1}i^2=\frac{(k+1)(k+2)(2k+3)}{6}}$

(e) Assume $\sum_{i=1}^{k}i^2=\frac{k(k+1)(2k+1)}{6}$, and show that $\sum_{i=1}^{k-1}i^2=\frac{(k+1)(k+2)(2k+3)}{6}$

I'm actually going to do an induction proof.

Base case: $n=1$ then $\sum_{i=1}^{1}i^2=1$

$\frac{n(n+1)(2n+1)}{6}=\frac{1(2)(3)}{6}=1$ LHS=RHS ✓

Induction Hypothesis: $\exists k\in\mathbb{N}$ s.t $\sum_{i=1}^{k}i^2=\frac{k(k+1)(2k+1)}{6}$

Induction Step: WTProve $\sum_{i=1}^{k+1}i^2=\frac{(k+1)(k+2)(2k+3)}{6}$

Note: $\sum_{i=1}^{k+1}i^2=\sum_{i=1}^{k}i^2+(k+1)^2$

$$\overset{(IH)}{=}\frac{k(k+1)(2k+1)}{6}+(k+1)^2$$

$$=\frac{k(k+1)(2k+1)+6(k+1)^2}{6}$$

$$=\frac{(k+1)[k(2k+1)+6(k+1)]}{6}$$

$$=\frac{(k+1)[2k^2+7k+6]}{6}$$

$$=\frac{(k+1)(k+2)(2k+3)}{6}$$ as desired.

8. Consider the sequence defined by $a_1=2$, $a_{n+1}=\frac{1}{2}(a_n+6)$. Which of the following statements is correct?

(a) $\{a_n\}$ is increasing and bounded above by 3
(b) $\{a_n\}$ converges to 5
(c) $\{a_n\}$ is increasing and bounded above by 5
(d) $\{a_n\}$ is increasing and bounded above by 6
(e) $\{a_n\}$ diverges

Try test points: $a_1=2$ $a_2=\frac{1}{2}(2+6)=4$

$a_3=\frac{1}{2}(4+6)=5$ So a_n is increasing...

Find proper upper bound: if a_n converges

then $\lim_{n\to\infty}a_{n+1}=\lim_{n\to\infty}a_n=L$

i.e. $\lim_{n\to\infty}\frac{1}{2}(a_n+6)=\frac{1}{2}(L+6)\overset{set}{=}L$

$\Rightarrow L+6=2L \Rightarrow 6=L$ Hence it approaches 6.

Answer: d)